STAT 412 QZ07 — Concise Answer Key

1. $H_0 : \mu = 162.2$; $H_1 : \mu \neq 162.2$. $z = 2.43$; P-value = 0.015. Reject H_0; sufficient evidence of a change. Reject for significance levels greater than the P-value.

2. Type I error probability = 0.0244. Type II error probability when $\mu = 178$ is 0.2266.

3. Choice B: $H_0 : p = 0.097$, $H_a : p < 0.097$. $z = -0.46$; P-value = 0.3225. Choice C: do not reject H_0; insufficient evidence to support the principal's claim.

4. Choice C.

5. $\chi^2 = 48.57$; P-value = 0.900. Not significant at 0.05; do not reject H_0; insufficient evidence that $\sigma^2 \neq 4.2$.

6. Choice F: $H_0 : \mu = 41$, $H_1 : \mu < 41$. $t = -2.46$; $0.0075 < P\text{-value} < 0.01$ (exact about 0.0082). Reject H_0; sufficient evidence that the mice live to be less than 41 months old.

7. Choice B: $H_0 : p = 0.5$, $H_a : p > 0.5$. $z = 7.75$; P-value = 0.000. Null-hypothesis conclusion: choice B. Final conclusion: choice D.

8. Minimum sample size required: $n = 20$ for each sample.